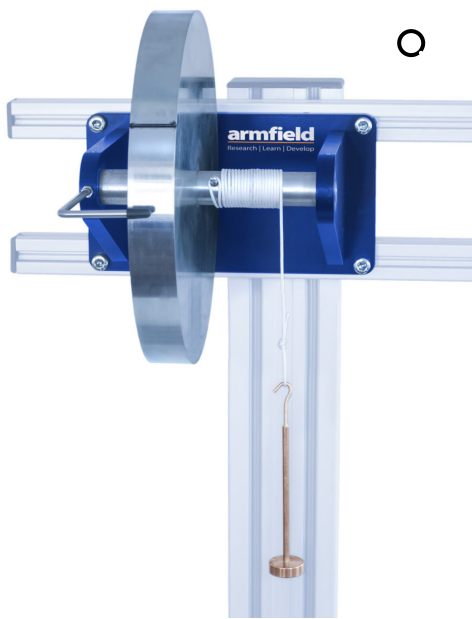




SD-1.55 – Simple Flywheel

Experiment to verify the second law of motion applied to a flywheel.

PRODUCT CODE: SD-1.55

Categories: Educational, SD-1.55-Simple Flywheel, SV - Theory Of Machines, SV Series - Statics & Vibrations

Tags: Angular Acceleration , Couple , Moment Of Inertia , Newton's Second Law Of Motion , Rotating Machinery , Rotational Acceleration , Rotational Mass , SD-1.55 , Simple Flywheel , Torque

DESCRIPTION

A steel disc Ø250mm diameter and 30mm thick is mounted on a shaft running in precision bearings housed in a substantial wall bracket. A mark on the flywheel and a pointer on the bracket enables the revolutions to be counted and timed with the stop watch supplied. A cord, load hanger and set of weights are provided to start the flywheel rotating. From the timings of a known number of revolutions, students can then verify the experimental capabilities of the apparatus.

TECHNICAL SPECIFICATIONS

- Coated Steel Flywheel
- **Flywheel:** Ø250 x 30mm thick
- **Weights set:** 10 x 0.1N, 10 x 0.2N, 1 x 1N, 2 x 2N, 1 x 5N, 1 x 10N
- 1 x Load hanger

FEATURES & BENEFITS

- Wall mounted apparatus
- Can be mounted to SV100 Bench mounted frame
- Flywheel 'locked' before weights applied
- All hangers and weights supplied
- Timing system supplied

Experimental Content

- To verify the second law of motion applied to a flywheel, i.e. the relationship between torque and angular acceleration
- To compare experimental and calculated moments of inertia of a disc

- To study the energy transformations and to demonstrate that a flywheel can be used to store energy

Ordering Specification

- 1 x SD-1.55 assembled
- 1 x Load hanger
- 10 x 0.1N, 10 x 0.2N, 1 x 1N weights
- 2 x 2N, 1 x 5N, 1 x 10N weights
- 1 x Stopwatch
- 1 x Tape measure
- Spare cord
- Instruction manual
- Packing list
- Test sheet

Series Includes:

Theory of Machines

- SD-1.50 Wheel and Axle
- SD-1.51 Wheel and Differential Axle Apparatus
- SD-1.52 Efficiency of Screw Threads
- SD-1.53 Gear Tooth Form Apparatus
- SD-1.54 Cam and Follower Apparatus
- SD-1.55 Simple Flywheel
- SD-1.56 Epicyclic Gear Train
- SD-1.57 Castor, Camber, and King Pin Inclination
- SD-1.58 Relation Between Angular and Linear Speeds
- SD-1.59 Disc Brake Apparatus
- SD-1.60 Static & Dynamic Balancing Apparatus
- SD-1.61 Governor Apparatus
- SD-1.62 Gear Assembly Unit Combined Drives
- SD-1.63 Single Epicyclic Gear Train

Essential Accessories

SV100: Bench Mounted Frame

Requirements

- **SV100:** Bench Mounted Frame
- Recommended suspension height not to exceed 1.0 metre

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

Volume: 0.014m³

Gross Weight: 25kg



Dimensions

Length: 0.19m

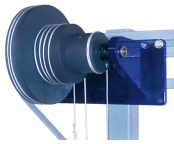
Width: 0.25m

Height: 0.30m

Order Codes

- **SD-1.55** Simple Flywheel
- **SV100** Bench Mounted Frame

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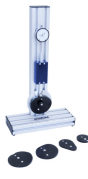
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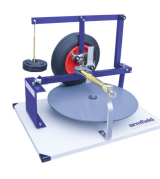
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